Credible Bank Resolution: Clearing, Runs, and Optimal Instruments

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Abstract

We solve the credible-resolution design problem of one fully specified two-bank institution with actual interbank payments, creditor runs, insured deposits, senior and designated debt, contractual AT1 losses and equity. Its bridge operations and temporary funding are legally specified functions, not design variables. Every legal continuous authority action is compared at every isolated, joint and off-path failure history. Paid-in capital varies over an interval, and stress and systemic probabilities each vary over the whole unit interval. An explicit finite candidate rule gives the global discretionary optimum; the commitment optimum has an exact classification, including flat boundary cases. Admitted lotteries cannot improve these values. With the creditor selected to wait at indifference, a particular stress probability yields a systemic-probability threshold of 15/49 and a strict commitment gap. We derive claim prices for the entire capital continuum, sharp observation-compatible sets, and two economies with identical realized prices and behavior but opposite credibility judgments for the same announcement. A general priority-clearing and real-algebraic framework treats complete legal domains, equilibrium pricing, empty feasibility, unbounded values and unattained suprema.

1 Original target, economic scope, and context

The original catalogue problem [1], entitled *Credible bank resolution*, fixes a banking system with insured deposits, unsecured claims of stated seniorities, equity and cross-bank exposures. A rule a specifies loss allocation, recapitalization, transfer of critical functions and temporary funding. Investors price claims before stress, anticipating the rule and actual authority deviations. The target is

$$\begin{aligned} A_{\mathrm{cred}} &= \{a \in A : V^A(a(s); s, a) \geq V^A(d; s, a) \text{ for all declared } s,\ d \in D(s)\}, \\ W_* &= \sup_{a \in A_{\mathrm{cred}}} W(a). \end{aligned} \tag{1.1}$$

The feasible set must follow the insolvency law, loss-absorbing claims and operational constraints. Welfare includes household surplus, payments and credit services, fiscal resources and spillovers. Runs and other banks' write-down losses must be modeled. Isolated and joint failures must both be assessed. Price-implied losses must be distinguished from sequential policy optimality, under explicit pricing and recovery restrictions.

The two named variants are an irrevocably enforceable loss allocation and a discretionary authority subject to all-state credibility. For a finite optimal value the requested output is a feasible ε-optimal rule for every $\varepsilon > 0$; exact attainment needs proof. Empty feasibility and unbounded

values are separate outcomes. The source's common conventions also require a declared equilibrium selection, complete budgets and clearing, common primitives across observations, and sharpness and endpoint claims. These conventions are used throughout.

This is a model-construction research agenda, not a numerical banking economy specified by the source. We therefore give both a general represented-institution theorem and a fully specified institution in which the entire rule domain is solved. Its legal domain is fixed before outcomes. The claim concerns that fixed-system task. It does not assert a theorem for every possible banking law, arbitrary non-effective utility, hidden-capital mechanism, or infinite strategic horizon.

The catalogue attributes the empirical motivation to Di Stefano, Lengwiler and Rishabh [2], and the broader policy agenda to Tucker [3]. The two earlier attempts [9, 10] already establish a one-bank quadratic spillover optimum and a multibank positive-definite exposure-cost characterization with observable-risk feedback. Their exposure costs do not determine payment clearing or runs, and their credible rule is a singleton at fixed primitives. Those results remain credited. The contribution here is the joint solution, in the fully specified institution of Section 5, of actual clearing, runs, a nontrivial credible instrument continuum and commitment, with a constructive sharp example and an explicit nonidentification pair for the associated identification questions.

2 Actual priority clearing

There are finitely many banks i and classes $r = 1, \dots, R_i$, ordered from most to least senior. At a fixed history and proposed resolution action, remaining face amounts are $b_{ir} \geq 0$. Protected deposits cannot be written down. Contractual AT1 absorption, designated resolution debt and senior debt have their specified distinct treatments. Write $\pi_{ir,j} \geq 0$ for the fraction of class (i, r) held by bank j, with $\sum_j \pi_{ir,j} \leq 1$; the residual is externally held. Outside assets and authorized bank cash are $e_i \geq 0$.

For a payment vector v in $\prod_{ir}[0, b_{ir}]$, define

$$\begin{aligned} A_i(v) &= e_i + \sum_{j,r} \pi_{jr,i} v_{jr}, \\ \Phi_{ir}(v) &= \min\left\{ b_{ir}, \left[A_i(v) - \sum_{s<r} b_{is} \right]_+ \right\}. \end{aligned} \tag{2.1}$$

A clearing vector satisfies $v = \Phi(v)$, and equity is $[A_i(v) - \sum_r b_{ir}]_+$. Payments within a class are proportional to ownership. Deposit-insurance transfers to external depositors are recorded outside this bank-cash equation. A payment legally entering another bank's assets must instead be included explicitly. The classical network-clearing context is Eisenberg–Noe [4]; the following proofs include priority classes and are supplied directly.

Theorem 2.1 (Unique clearing and certified errors)**.** *Suppose $\sum_j \pi_{ir,j} \leq 1 - \eta$ for every class, for some $\eta > 0$. There is exactly one clearing vector v^*. Simultaneous iteration converges from every vector in the payment box, with*

$$\|v^t - v^*\|_1 \leq (1-\eta)^t \|v^0 - v^*\|_1, \qquad \|v - v^*\|_1 \leq \eta^{-1} \|v - \Phi(v)\|_1.$$

For fixed faces and ownership fractions,

$$\|v^*(e) - v^*(\widetilde{e})\|_1 \leq \eta^{-1} \|e - \widetilde{e}\|_1.$$

Proof. For one bank, let $w_i(a)$ be its waterfall vector as a function of resources $a \geq 0$. Each coordinate is nondecreasing and $\sum_r w_{ir}(a) = \min\{a, \sum_r b_{ir}\}$. Therefore

$$\|w_i(a) - w_i(a')\|_1 = \left| \min\{a, \sum_r b_{ir}\} - \min\{a', \sum_r b_{ir}\} \right| \leq |a - a'|.$$

Consequently

$$\begin{aligned}\|\Phi(v)-\Phi(v')\|_1 &\le \sum_i \left|\sum_{j,r}\pi_{jr,i}(v_{jr}-v'_{jr})\right| \\ &\le \sum_{j,r}\left(\sum_i \pi_{jr,i}\right)|v_{jr}-v'_{jr}| \le (1-\eta)\|v-v'\|_1.\end{aligned}$$

The map preserves the closed payment box. Its successive iterate differences form a summable geometric bound, so iterates converge to a fixed point by continuity. Two fixed points have distance at most $1-\eta$ times their distance and hence coincide. Comparing any iterate with that fixed point gives the first bound. The triangle inequality gives $\|v-v^*\|_1 \le \|v-\Phi(v)\|_1 + (1-\eta)\|v-v^*\|_1$, which proves the residual bound. Applying the same inequalities to maps with different e, and moving the contraction term to the left, proves the final assertion. □

If fractions also change while faces stay fixed, the asset-difference bound before division by η acquires the additional term

$$\sum_{j,r}\widetilde{v}^*_{jr}\sum_i|\pi_{jr,i}-\widetilde{\pi}_{jr,i}|.$$

This follows by inserting the second fixed point into the first map. The first map's leakage bound suffices. Varying faces is not covered without extra terms.

Theorem 2.2 (Clearing without leakage)**.** *Under merely $\sum_j \pi_{ir,j} \le 1$, least and greatest clearing vectors exist. They have effective semialgebraic graphs for represented semialgebraic parameters. Either may be a declared clearing convention; uniqueness and the preceding geometric bounds are not asserted.*

Proof. The map Φ is continuous and coordinatewise nondecreasing. Iteration from the full face vector b decreases and is bounded below by zero. Coordinate limits exist and form a fixed point v^+. Every fixed point z satisfies $z \le b$, hence $z \le \Phi^t(b)$ for every t, and therefore $z \le v^+$. Iteration from zero similarly gives the least fixed point v^-.

Let $C(v)$ denote the box constraints and fixed-point equations (2.1). The greatest graph is

$$C(v) \ \wedge \ \forall z\,\big(C(z) \Longrightarrow z \le v\big);$$

the least graph reverses the final inequality. The min and positive-part operations have finite polynomial descriptions, so real quantifier elimination gives the parameter assertion.

There is also a direct exact algorithm at rational input. Each coordinate of Φ is zero, its full face, or resources minus senior faces. Enumerate all branch patterns and solve the resulting bounded linear polyhedra. Their union is exactly the clearing set, including boundary overlap. In the unique case every feasible branch gives the same vector. In general maximize or minimize $\sum_{ir} v_{ir}$ over every nonempty branch and compare its optima. Since v^+ dominates every clearing, it is the unique maximizer of this sum; v^- is the unique minimizer. Thus the procedure is finite even if monotone iteration never terminates in finitely many steps. □

Selecting greatest or least clearing is an economic convention, not a uniqueness conclusion. An all-clearing welfare criterion must retain the whole correspondence C in its quantifiers. The concrete institution below has leakage and direct acyclic clearing, so this selection issue does not arise there.

3 Complete legal domains, incentives, and exact design

3.1 The represented institution

The following are input assumptions for a general theorem, not claims about every real resolution regime. Coefficients are rational or explicitly represented real algebraic numbers. All stated sets and function graphs are given by finite real polynomial formulas.

A finite perfect-recall prefix game includes any specified bank financing or risk decisions, Nature signals and creditor withdrawal actions. Each terminal prefix history reaches one authority decision and then the specified passive settlement. The authority's public history contains all its payoff-relevant information. If unresolved private uncertainty is present, its conditional beliefs must instead be explicit in the model; they cannot silently be replaced by a realized public state.

The legal infrastructure parameter k ranges over a supplied semialgebraic set. A finite declared list $\mathcal{H}$ of failure/run histories is fixed independently of their equilibrium probabilities. At each h, the entire nonempty legal action set is $D_h(k)$. It may be a finite union of components for discrete choices and full continuous intervals for cash, transfers and write-downs. A complete rule is $a = (k, (x_h)_{h\in\mathcal{H}})$ subject to every initial-finance, operational and legal constraint; their conjunction defines A.

Post-resolution clearing is the unique vector of Theorem 2.1, an explicitly selected extreme vector of Theorem 2.2, or another proved represented convention. Denote it by $v^*(k, h, x)$. All passive recoveries, temporary funding repayments, resources and claimant payoffs are specified. The authority value $V_h(k, x, v^*)$ and private payoffs are finite semialgebraic functions. At each fixed announced rule they are uniformly bounded on the admitted equilibrium/price domain, but no common bound over all rules is required. A later strategic authority stage would require its own continuation constraints; terminal settlement is an explicit assumption.

Define

$$\begin{aligned}\mathrm{Cred}(a) \iff &\text{for every } h \in \mathcal{H} \text{ and } d \in D_h(k),\\ &V_h(k, x_h, v^*(k, h, x_h)) \geq V_h(k, d, v^*(k, h, d)).\end{aligned} \tag{3.1}$$

This recomputes clearing on the deviation's own balance sheet. Using the announced rule's clearing for a different haircut would be incorrect. Since there is one terminal authority decision, (3.1) compares its complete continuation deviations and is precisely the original credibility condition in this institution. Its graph is semialgebraic by substituting the clearing and payoff graphs and eliminating the deviation variables. No differentiability or first-order sufficiency is used.

If legally admitted rules randomize, their full representation or a proved reduction is needed. Deterministic rules cannot silently replace them. Section 9 proves such a reduction for all the lotteries admitted in our concrete institution.

3.2 A sequential-equilibrium graph

For a fixed rule, Nature law and fixed prices, prune only histories impossible because of Nature. Do not delete a legally available strategic action because its equilibrium probability is zero. Write σ for behavioral probabilities and μ_I for beliefs over the nodes of an information set I. Let $R_h(\sigma)$ be the product of chance and behavior probabilities reaching node h, and $R_I(\sigma) = \sum_{h\in I} R_h(\sigma)$.

Consistency in the sense of Kreps–Wilson [5] is exactly the formula

$$\begin{aligned}\forall \epsilon > 0\ \exists \sigma', \mu' :\ &\sigma' \text{ completely mixed}, \quad \|\sigma' - \sigma\|_\infty < \epsilon, \quad \|\mu' - \mu\|_\infty < \epsilon,\\ &R_I(\sigma') > 0, \qquad \mu'_I(h) R_I(\sigma') = R_h(\sigma') \quad (h \in I, \text{ every } I).\end{aligned} \tag{3.2}$$

All probability vectors belong to their simplices. The same σ' witnesses every information set,

and the Nature parameters are held fixed inside this formula. One implication is immediate from a consistent sequence; conversely witnesses at $\epsilon = 1/n$ form the required sequence. Full mixing gives positive R_I precisely on the pruned game. When Nature probabilities depend on parameters, use their finitely many support strata first. This pruning never deletes a declared authority history from the separate quantifier in (3.1).

Let $U_i(I; \sigma, \mu_I)$ be player i's conditional continuation payoff. For each of its finitely many pure continuation plans τ_i, impose

$$U_i(I; \sigma, \mu_I) \geq U_i(I; \tau_i, \sigma_{-i}, \mu_I). \tag{3.3}$$

These are finite polynomial comparisons after introducing the payoff graphs. Perfect recall means no information set repeats along a path, so the continuation expression is affine in each deviating behavioral coordinate separately. Maximizing such an expression over the product of simplices has a pure-plan maximizer, by maximizing one coordinate at a time. Thus (3.3) tests all randomized continuation deviations. Together (3.2)–(3.3) give an exact finite real formula for sequential assessments, not arbitrary off-path beliefs.

Lemma 3.1 (Existence for the fixed finite prefix)**.** *A finite perfect-recall prefix game with fixed terminal payoffs has a sequential equilibrium after the stated Nature pruning.*

Proof. If there are no strategic information sets, the empty assessment proves the assertion. Otherwise proceed as follows. Fix $\delta > 0$ smaller than the reciprocal of every information-set action count m_I. Replace each behavioral simplex by $\sigma_I(a) = \delta + (1 - m_I\delta)z_I(a)$, with z_I in a simplex. Treat every information set as an agent whose payoff is the original player's ex ante payoff. This is a finite agent normal-form game: fixing an agent's pure choice determines the displayed noisy action, and independent mixing of agents gives exactly these behavioral probabilities. The finite-game Nash theorem [6, Theorem 1] gives an agent equilibrium.

Every remaining information set has positive reach. Perfect recall makes its reach independent of that agent's current action and makes its conditional node probabilities invariant to the original player's earlier behavior once that information set is reached: the player's own past action factors are common to its nodes and cancel. Consequently each agent best response is a conditional best response among floor-constrained actions. Backward replacement, starting at the last information sets, shows that no floor-constrained continuation plan improves at any information set. At a later information set the same cancellation of the player's past factors preserves the conditional distribution during this replacement. This proves the full continuation comparison.

Take a sequence $\delta \downarrow 0$ and a convergent subsequence of the behavior and Bayes-belief vectors, which lie in a finite product of compact simplices. It is consistent because each entire profile is completely mixed with the same fixed Nature law. For any pure continuation plan, approximate it by putting mass δ on each other action and the remaining mass on its prescribed action. The floor-constrained comparison just proved passes to the limit: conditional payoff expressions are polynomials in the behavior and belief coordinates. It gives (3.3) for every pure continuation plan at every information set. There are finitely many such comparisons. Hence the limit is a sequential equilibrium. □

On $\mathrm{Cred}(a)$, an indifferent authority honors its prescribed optimal continuation as an explicit selection convention. If the target instead ranges over all optimal authority continuations, include all of them in the equilibrium graph. Let $\mathcal{E}(a)$ contain the prefix assessment, all declared budgets, and competitive prices

$$Q_c = \mathbb{E}[MZ_c]$$

for the specified common pricing kernel and actual payoff Z_c. If prices enter private incentives they enter those same payoff functions. Lemma 3.1 does not prove nonemptiness after imposing

these extra price-consistency equations. Nonemptiness is separately proved in the concrete model, and is tested in the general graph.

3.3 The global design theorem

A selected-equilibrium target includes its represented selection graph. Alternatively, declare robust-prefix welfare $W_-(a) = \inf_{e\in\mathcal{E}(a)} W(a,e)$ on nonempty $\mathcal{E}(a)$. The exact graph of its finite value w is

$$\begin{aligned} &\forall e \in \mathcal{E}(a), \quad W(a,e) \geq w, \\ &\forall \epsilon > 0 \quad \exists e \in \mathcal{E}(a), \quad W(a,e) < w + \epsilon. \end{aligned} \tag{3.4}$$

The second condition matters when the infimum is not attained. This robust convention is a declared target, not an implicit choice imposed on the original question.

Theorem 3.2 (Exact design over the entire represented domain)**.** *For the represented institution and either a represented selected welfare or* (3.4)*, there is an exact finite algorithm for the feasible credible set, its welfare value set and supremum. It distinguishes empty feasibility, a finite supremum and an unbounded value. In the finite case it decides attainment and returns an exact optimizer when attained. For every specified positive rational or algebraic ϵ, it returns an algebraic feasible credible rule with welfare greater than $W_* - \epsilon$. The corresponding existence statement holds for every real $\epsilon > 0$. No polynomial running-time claim is made.*

Proof. Conjoin $a \in A$, $\mathrm{Cred}(a)$, $\mathcal{E}(a) \neq \varnothing$, and the welfare graph. Everything is a finite real formula by (2.1)–(3.4). Effective real quantifier elimination and real-algebraic sampling [7, 8] apply: their inputs are finitely many polynomial conditions with represented algebraic coefficients; nonpolynomial semialgebraic functions have been replaced by their graphs. The output describes the exact projected value set $S \subset \mathbb{R}$, a finite union of intervals and points with algebraic finite endpoints.

Ordinary real-algebraic sampling also follows constructively from this elimination operation. Project a nonempty represented graph to one coordinate. Univariate real-root isolation describes its nonempty intervals and isolated algebraic points; choose a rational interior point of an interval, or an isolated algebraic point. Fix that coordinate and repeat on the nonempty fiber. After finitely many coordinates this gives an ordinary real-algebraic witness, not merely a point in an infinitesimal extension.

If S is empty, feasibility fails. Otherwise its upper endpoint is algebraic or $+\infty$. For a finite endpoint s, membership of s decides attainment, and sampling the original graph at that value returns an optimizing rule and any required witnesses. If it is not attained, the defining property of the supremum makes the graph with $W > s - \epsilon$ nonempty. Algebraic sampling again gives the legal rule, not only a numerical lower bound on its value. For an arbitrary positive real tolerance, choose a smaller positive rational tolerance to obtain existence. If the value is unbounded, this is reported separately. □

These real-algebraic tools are established results. Their content used here is effective elimination of finite real quantifiers and extraction of algebraic sample points; see [8, Theorem 2.27] and the sampling construction above. The many continuation plans and quantifier blocks may be expensive. This manuscript does not claim to have executed a general CAD or quantifier-elimination computation.

4 What common-kernel claim prices identify

Fix a complete finite settlement-state list $s = 1, \dots, L$, including normal repayment. If only stress states are retained, explicitly subtract the known normal-state price contribution. Let R_{sc} be the known actual payoff of claim c, incorporating runs and clearing. For this lemma the physical probabilities $\pi_s > 0$ are known and a common positive pricing kernel obeys known finite bounds $0 < \ell_s \le M_s \le u_s$. Put $q_s = \pi_s M_s$ and define

$$\mathcal{D}(Q) = \{q : R^\top q = Q, \quad \pi_s \ell_s \le q_s \le \pi_s u_s \ \forall s\}. \tag{4.1}$$

A safe unit-payoff claim, if observed, is a column of R and fixes $\sum_s q_s$.

Theorem 4.1 (Sharp state-price information)**.** *$\mathcal{D}(Q)$ is exactly the identified state-price set. It is empty precisely for incompatible prices and restrictions. If nonempty, the sharp bounds on $t^\top q$ are its attained linear-program extrema. That target is point identified exactly when*

$$t \perp \operatorname{span}(\mathcal{D}(Q) - \mathcal{D}(Q)).$$

If a point of $\mathcal{D}(Q)$ is strictly inside every box bound, this is equivalent to t belonging to the row space of $R^\top$.

Proof. Every admitted kernel gives (4.1). Conversely a retained q gives $M_s = q_s/\pi_s$, satisfying the bounds and every price equation. This proves sharpness. The set is closed and bounded, so the linear extrema are attained. A linear functional is constant exactly when it annihilates every feasible difference. At a strictly interior point q_0, every $v \in \ker R^\top$ has $q_0 + tv$ feasible for all sufficiently small $|t|$. Thus the difference span is the kernel, whose orthogonal complement is the row space. □

Binding kernel bounds can identify a target without the row-space criterion, so the interior qualification cannot be dropped. If physical probabilities are unknown and null states are allowed, the sharp joint set instead uses

$$\pi \ge 0, \quad \sum_s \pi_s = 1, \quad R^\top q = Q, \quad \ell_s \pi_s \le q_s \le u_s \pi_s.$$

On a null state these inequalities force $q_s = 0$, and any in-bound M_s can be assigned there; on a positive state use $M_s = q_s/\pi_s$. This also proves the converse. If strictly positive support is required, replace $\pi \ge 0$ by $\pi > 0$; compactness may fail and endpoints must be reported as infima/suprema with their actual attainment flags.

Even complete state prices do not separate an unknown physical law from its kernel or reveal the authority's ranking of unused legal deviations. For a represented structural class, impose all observed prices, run and payment probabilities, and parameter restrictions in one common graph. Projection gives sharp joint sets for the credibility indicator, optimal instruments and welfare. An announced rule and actual discretionary actions are separate variables: their equality must be tested, not assumed in order to infer credibility. Section 10 supplies both a constructive sharp example and an explicit nonidentification pair.

5 A fully specified banking institution

5.1 Balance sheets, timing, and funded capital

Fix banks A, B and put $\gamma = 1/2$. Bank A has insured deposits of face 1, protected senior unsecured debt of face 1/2, designated resolution debt of face 1, contractual AT1 of face 1/4,

and old equity of 1/4. In normal conditions its assets 3 pay all debt and leave old equity 1/4. In stress its pre-resolution assets are 3/2: old equity is wiped and AT1 contractually absorbs 1/4. The remaining debt faces total 5/2, leaving recapitalization shortfall 1. AT1 absorption is not a discretionary resolution haircut.

Bank B holds fraction γ of A's designated debt; its other half and all other claims are held outside the bank network. An A haircut x therefore reduces B's actual payment receipt by γx. This is an interbank asset, not an assumed spillover cost.

At date zero the designer publicly chooses paid-in capital $K \in [0, 1]$. It contributes cash γK to B before stress and has a real capital-raising/locking friction $K/2$. Cash principal is a financial contribution with residual ownership, not another social loss. Adequately funded outside owners supply it through a legally binding date-zero capital assessment. They retain the stated residual ownership; voluntary participation or withdrawal is not an additional unmodeled decision. The contribution is not a promise of future rescue and is locked before the terminal authority acts. Any public lottery over K is drawn before and independently of the subsequent Nature shocks, with no preliminary signal about them. In particular, the conditional shock law below remains (p, ρ) given its realization.

Bank B has insured deposits γ and designated wholesale debt γ. In a stress state H, its external assets are $\gamma(1 + K - H)$. With the actual incoming A payment they become

$$\gamma(1 + K - H) + \gamma(1 - x) = \gamma(2 + K - H - x). \tag{5.1}$$

The declared states are $H = 0$, an isolated initial loss at A, and $H = 1/2$, an additional external-asset loss at B. At $K = 0$ the latter makes both banks initially insolvent. Capital and incoming write-downs endogenously change subsequent shortfalls. In normal conditions $H = x = 0$, B's faces are fully paid and equity is γK.

Nature generates stress with probability $p \in [0, 1]$; conditional on stress, $H = 1/2$ has probability $\rho \in [0, 1]$. Normal claims are fully paid. Both hypothetical stress states and both creditor histories remain in the declared credibility list even at probability endpoints. The probability square is an exogenous primitive family, independent of capital choices. At $p = 0$, ρ labels counterfactual stress composition rather than an observable positive-probability conditional law.

5.2 Runs, temporary liquidity, and the entire legal domain

After observing K, H but before resolution, the outside wholesale creditor chooses *wait* or *run*. Waiting retains its designated claim. A run exercises early redemption: a standing central-bank facility lends γ to B, which pays γ to a settlement channel. The creditor receives $\gamma/2$; the remaining $\gamma/2$ is a real processing/fire-sale cost. Bank cash changes by $+\gamma - \gamma = 0$, and the wholesale liability is replaced by a protected central-bank liability γ. Its pre-resolution assets (5.1) are unchanged. The advance is repaid after recapitalization; it is not counted again as an unrecovered fiscal loss. The standing advance is a binding right of this fixed institution under both variants. Withholding it is not a legal terminal deviation.

A run also incurs real critical-service interruption 7/20. Insured deposits and A's senior unsecured debt are protected. Required payment functions continue through the statutory bridge operation; liquidation without those functions is outside the stated operational law. For waiting, the authority's *entire* legal action set is

$$0 \le x \le 1, \qquad 0 \le y \le \min\{1, [H + x - K]_+\}. \tag{5.2}$$

Here x is A's designated haircut and γy is B's. A solvent bank's equity cannot remain while its designated debt is impaired. All old insolvent equity is wiped. The minimum required public bank cash is

$$b_A = 1 - x, \qquad b_B = \gamma\big([H + x - K]_+ - y\big). \tag{5.3}$$

Excess cash gifts beyond restoration of remaining debt faces are legally disallowed; they are not an omitted authority action.

If $H+x-K>0$, B's assets plus b_B are $\gamma(2-y)$, exactly paying insured deposits γ and wholesale debt $\gamma(1-y)$. Otherwise $b_B=y=0$, all faces are paid, and equity is $\gamma(K-H-x)$. Bank A's estate is $3/2+b_A=5/2-x$, exactly paying its protected $3/2$ and designated $1-x$. Thus the actual interbank payment and both priority waterfalls clear. These identities also prove clearing directly without iteration.

After a run $y=0$, because the replacement central-bank claim is protected, and *every* $x\in[0,1]$ remains a legal deviation. Formula (5.3) restores the needed estate in that branch too. Bridge operating purchases cost $x/4$. A stress-date cash capacity of 2 covers all legal actions: $b_A+b_B\le 5/4$, the advance is at most $1/2$, and operating purchases are at most $1/4$. The advance counts toward peak liquidity, although it is later repaid. Operating inputs are purchased from a prefunded reserve at their real opportunity cost. Date-zero capital cash and its raising friction are separate date-zero flows. Outside household/taxpayer cash may be fixed at 10, so none of these finite funding requirements relies on an unavailable resource.

The complete announced rule specifies the action in (5.2) at both waiting states and an x after each run, as well as K. Recapitalization and bridge/temporary-funding operations are the legally specified functions just given. Thus no feasible resolution action has been discarded by representing the rule this way. Arbitrary legal action lotteries are addressed in Section 9.

5.3 Credit production, private payoffs, and welfare

Wholesale receipts fund a separate working-capital sector through a pass-through par lending facility. With normalized funds $z\in[0,1]$, entrepreneurs choose $0\le q\le z$, use working capital γq, and produce gross resources $\gamma q+q-q^2/2$. The principal γq is repaid at par. Net real surplus

$$F(q)=q-q^2/2$$

is maximized at $q=z$. Entrepreneurs own that surplus; the creditor's financial payoff remains its cash repayment, subsequently lent and returned at par. With unit discounting, this pass-through use does not silently add entrepreneurial surplus to the debt price or run incentive. Normal $z=1$ gives $F(1)=1/2$. Waiting gives $z=1-y$, so the production loss is $y^2/2$. Running gives $z=1/2$, so it is $1/8$.

The social numeraire is the real consumption good, with common unit weights on its owners' resources. Public contributions b_A,b_B have real fiscal distortions b_A and $2b_B$, respectively; principal transfers are not counted a second time. Prefunded bridge operating inputs have total real cost $x/4$. Conditional on waiting, the authority minimizes precisely

$$L_H(x,y;K)=1-x+[H+x-K]_+-y+x/4+y^2/2. \tag{5.4}$$

After running its action-dependent loss is (5.4) at $y=0$. The additional loss fixed by that branch is

$$R=\gamma/2+7/20+1/8=29/40. \tag{5.5}$$

This includes processing, service interruption and the resulting production loss. It cannot be reversed by the authority's x choice but is included in ex ante welfare.

All other shock losses and operating surplus are common across designs at fixed (p,ρ). Their baseline $W_0(p,\rho)$ may depend on the shock law. Thus

$$W(K;p,\rho)=W_0(p,\rho)-J(K;p,\rho).$$

The baseline cancels in matched policy/institution comparisons at the same primitives, not automatically across distinct laws.

The creditor receives $\gamma(1-y)$ after waiting and $\gamma/2$ after running. It waits when the anticipated waiting haircut is at most $1/2$. At indifference select waiting. This is an explicit selected-equilibrium convention, not a claim that all indifferent creditors behave identically. Existing claims trade between adequately funded outside households; price is their valuation, not an additional unmodeled issuer-financing constraint. Initial capital funding was separately supplied above.

6 Every authority deviation and the endogenous run

Theorem 6.1 (The complete discretionary continuation)**.** *For every $K \in [0,1]$ and $H \in \{0, 1/2\}$, the unique authority optimum after waiting is*

$$x_W = \min\{1, K - H + 3/4\}, \qquad y_W = \min\{3/4, 1 + H - K\}. \tag{6.1}$$

After a run the unique optimum is

$$x_R = [K - H]_+, \qquad y_R = 0. \tag{6.2}$$

With the stated creditor selection, the run rule is

$$H = 0 : \ K < 1/2; \qquad H = 1/2 : \ K < 1.$$

The rule is credible at both isolated and joint states and at both creditor histories, independently of their probabilities.

Proof. Fix a feasible y. If $y > 0$, legality requires $x \geq K - H + y$. Above the resource kink the part of the loss varying with x has slope $1/4 > 0$. If $y = 0$ and $K > H$, it has slope $-3/4$ up to $x = K - H$ and $1/4$ thereafter. If $K \leq H$, the kink is at or below zero. In all cases the unique minimizing x is

$$x^*(y) = [K - H + y]_+, \qquad 0 \leq y \leq \min\{1, 1 + H - K\}.$$

Substitution into (5.4) gives the strictly convex function

$$f_H(y; K) = 1 + H - K - y + \frac{[K - H + y]_+}{4} + \frac{y^2}{2}. \tag{6.3}$$

Before the kink its derivative is $y - 1$; after it the derivative is $y - 3/4$. Since $H - K \leq 1/2$, the derivative is still negative through any nonnegative kink. The minimum is therefore $y = 3/4$ unless the upper endpoint $1 + H - K$ intervenes. This proves (6.1); the lower x bound is automatic because $K - H + 3/4 \geq 1/4$. Strict convexity in y, and uniqueness of $x^*(y)$, prove uniqueness over the entire legal set.

After running $y = 0$; the same two slopes give (6.2), whose upper bound is valid since $K - H \leq 1$. The fixed loss R does not affect this minimization. Finally the cash comparison says run exactly when $y_W > 1/2$. For $H = 0$, (6.1) crosses that level at $K = 1/2$; for $H = 1/2$ it crosses at $K = 1$. The equalities use the declared wait selection. This proves all continuation comparisons, including off-path histories. □

The proof gives feasible behavior and optimal continuations directly, so the concrete equilibrium exists. Before stress, competitive risk-neutral valuation assigns the expected actual repayments derived below. Those prices do not feed an additional financing decision, so there is no unsolved price-consistency fixed point.

7 The global credible capital optimum

Write $S_0(K)$ and $S_J(K)$ for the conditional real stress losses, including R when a run occurs. Direct evaluation of (6.1)–(6.2) gives

$$S_0(K) = \begin{cases} 69/40 - 3K/4, & 0 \le K < 1/2, \\ 1/4 + (1-K)^2/2, & 1/2 \le K \le 1, \end{cases}$$
$$S_J(K) = \begin{cases} 89/40 - K, & 0 \le K < 1/2, \\ 21/10 - 3K/4, & 1/2 \le K < 1, \\ 3/8, & K = 1. \end{cases} \tag{7.1}$$

For example, in the single run state $x_R = K$, so the variable loss is $1 - 3K/4$; adding $29/40$ gives the first line. In the joint run state $x_R = 0$ below $1/2$, giving $3/2 - K + 29/40$; above $1/2$, $x_R = K - 1/2$, giving $3/2 - K + (K - 1/2)/4 + 29/40$. In a single waiting state $x_W = 1, y_W = 1 - K$; in the joint waiting state at $K = 1$, $x_W = 1, y_W = 1/2$. These give all remaining lines and prove (7.1) without a numerical optimization.

The ex ante loss is

$$J(K; p, \rho) = K/2 + p\big((1-\rho)S_0(K) + \rho S_J(K)\big). \tag{7.2}$$

Equivalently,

$$J(K; p, \rho) = \begin{cases} p(69/40 + \rho/2) + K[1/2 - p(3+\rho)/4], & 0 \le K < 1/2, \\ p(3/4 + 27\rho/20) + K(1/2 - p + p\rho/4) \\ \qquad + p(1-\rho)K^2/2 & 1/2 \le K < 1, \\ 1/2 + p(1/4 + \rho/8), & K = 1. \end{cases} \tag{7.3}$$

At $1/2$ the value drops by $39p(1-\rho)/40$ relative to its left limit; at 1 it drops by $39p\rho/40$. These jumps may vanish at probability endpoints, but never point upward.

Theorem 7.1 (Exact design for the whole probability square)**.** *For every $(p, \rho) \in [0,1]^2$, define*

$$\mathcal{C}(p, \rho) = \{0, 1/2, 1\}$$

and adjoin

$$K_c = \frac{p(1 - \rho/4) - 1/2}{p(1-\rho)} \tag{7.4}$$

exactly when $p > 0, \rho < 1$ and $1/2 < K_c < 1$. A least value of (7.3) on this finite set is the global minimum over all $K \in [0,1]$, and its associated rule in Theorem 6.1 is an exact welfare-optimal credible rule.

Proof. The first branch is affine. Its infimum is at zero or its right limit, and the actual value at $1/2$ weakly improves that limit. The middle branch is convex quadratic. When its curvature is positive, its only possible strict interior minimum is the zero of its derivative, (7.4); otherwise it is affine or constant. Its endpoint possibilities are $1/2$ and the left limit at 1, which the actual value at 1 weakly improves. This proves the candidate rule even at $p = 0$ or $\rho = 1$, without dividing by zero, and proves attainment despite selected-equilibrium jumps. A minimizing candidate suffices; the assertion does not need to enumerate every point of a possible flat branch. □

Corollary 7.2 (The systemic threshold at $p = 1/2$)**.** *With the wait-at-tie creditor selection of Section 5, when $p = 1/2$ the optimizer is uniquely $K = 1/2$ for $\rho < 15/49$, uniquely $K = 1$ for $\rho > 15/49$, and exactly the two points $\{1/2, 1\}$ at equality.*

Proof. The first branch becomes $69/80 + \rho/4 + K(1-\rho)/8$, and the middle derivative is $\rho/8 + (1-\rho)K/2 > 0$. Thus only $0, 1/2, 1$ need be compared. Their exact differences are

$$\begin{aligned} J(0) - J(1/2) &= 17(1-\rho)/40, \\ J(0) - J(1) &= 19/80 + 3\rho/16, \\ J(1/2) - J(1) &= (-15 + 49\rho)/80. \end{aligned}$$

For completeness, $J(1/2) = 7/16 + 27\rho/40$ and $J(1) = 5/8 + \rho/16$. The only flat first branch is $\rho = 1$, whose value is still strictly above $J(1)$. Hence there are no additional optimizers. □

Corollary 7.3 (Rectangular ambiguity in incremental costs)**.** *At fixed K, J is nondecreasing in both p and ρ. For a declared incremental-cost criterion and uncertainty rectangle $[p_\ell, p_u] \times [\rho_\ell, \rho_u]$, its worst case is (p_u, ρ_u). Theorem 7.1 at that corner is an exact minimax incremental-cost design.*

Proof. Both conditional losses are nonnegative. Their difference is $1/2 - K/4 > 0$ below $1/2$, $27/20 + K/4 - K^2/2 > 0$ on $[1/2, 1)$, and $1/8$ at 1. The middle expression is concave and positive at both endpoints, which proves its stated sign. Formula (7.2) proves monotonicity. The same upper corner is a worst case for every K, so minimizing there gives the minimax rule. A claim about total welfare across laws would also have to include $W_0(p, \rho)$; this corollary does not silently cancel that baseline. □

8 Irrevocable commitment on the same legal domain

In the commitment variant the chosen loss allocation is irrevocably implemented. The balance sheets, protected claims, standing liquidity facility, capital instrument and resource constraints do not change. To induce waiting a deterministic committed rule must have $y \le 1/2$.

Theorem 8.1 (Conditional commitment and its global optimum)**.** *With the wait-at-tie creditor selection of Section 5, for every K, H a best committed rule induces waiting. Its statewise actions and losses are*

state and range	(x, y)	*stress loss*
$H = 0,\ K \le 1/2$	$(K + 1/2, 1/2)$	$3/4 - 3K/4$
$H = 0,\ K \ge 1/2$	$(1, 1 - K)$	$1/4 + (1-K)^2/2$
$H = 1/2,\ 0 \le K \le 1$	$(K, 1/2)$	$9/8 - 3K/4.$

(8.1)

The global loss is

$$J_{\mathrm{com}}(K; p, \rho) = \begin{cases} p(3/4 + 3\rho/8) + K(1/2 - 3p/4), & 0 \le K \le 1/2, \\ p(3/4 + 3\rho/8) + K(1/2 - p + p\rho/4) & \\ \qquad + p(1-\rho)K^2/2 & 1/2 \le K \le 1. \end{cases} \tag{8.2}$$

Its complete set of minimizers is

$$\begin{cases} \{0\}, & p < 2/3, \\ [0, 1/2], & p = 2/3,\ \rho < 1, \\ [0, 1], & p = 2/3,\ \rho = 1, \\ \{1/2\}, & p > 2/3,\ p \le 2/(2+\rho), \\ \{1\}, & p > 2/3,\ p > 2/(2+\rho),\ p\rho \ge 2/3, \\ \{K_c\}, & p > 2/3,\ p > 2/(2+\rho),\ p\rho < 2/3. \end{cases} \tag{8.3}$$

The last case always has $p > 0, \rho < 1$ and $1/2 < K_c < 1$.

Proof. The reduction (6.3) applies to every legal waiting action. For $y \le 1/2$ its derivative is negative on either side of the kink. Thus the best feasible y is $\min\{1/2, 1 + H - K\}$, and $x = [K - H + y]_+$. Substitution gives (8.1), including the shared boundary $K = 1/2$.

Inducing a run cannot improve this feasible waiting commitment. For the single state below $K = 1/2$, the least run loss exceeds the displayed waiting loss by $39/40$. For the joint state below $1/2$, the difference is $11/10 - K/4 \ge 39/40$; at and above $1/2$ it is $39/40$. These follow by subtracting (8.1) from the minimized run loss in (6.2) plus R. In the single state above $1/2$, the unconstrained waiting optimum already has $y \le 1/2$. Its loss is no larger than the minimum of the $y = 0$ subset, and a run adds the positive constant R. This argument also bounds any hypothetical run at a parameter where the wait-at-tie rule makes it impossible to induce. A nonminimizing run allocation cannot do better.

Taking the state expectation and adding $K/2$ proves (8.2). The two formulas agree at $1/2$. The left derivative is $1/2 - 3p/4$. The right derivative at $1/2$ is $1/2 - p(2+\rho)/4$, larger by $p(1-\rho)/4$, and thereafter increases with slope $p(1-\rho)$. At 1 it equals $1/2 - 3p\rho/4$. Hence the entire function is convex.

If $p < 2/3$, all derivatives are positive and the optimum is zero. At $p = 2/3$, the left interval is flat; the right derivative is positive unless $\rho = 1$, when the whole interval is flat. For $p > 2/3$, the left derivative is negative. A nonnegative right derivative at $1/2$ is exactly $p \le 2/(2+\rho)$, giving that endpoint. Otherwise a nonpositive derivative at 1 is exactly $p\rho \ge 2/3$, giving 1. In the remaining case the derivative has its unique zero K_c strictly inside the right interval. That case cannot have $\rho = 1$; its curvature is positive. At endpoint derivative equalities with $\rho < 1$, positive curvature prevents an additional flat interval. This proves every case in (8.3). □

Corollary 8.2 (A strict commitment comparison)**.** *With the wait-at-tie creditor selection of Section 5, at $p = 1/2$ commitment has unique optimum $K = 0$, with*

$$J^*_{\text{com}} = 3/8 + 3\rho/16.$$

Its waiting rules are $(x, y) = (1/2, 1/2)$ in the isolated state and $(0, 1/2)$ in the joint state. If J^*_{cred} denotes the discretionary minimum, then

$$J^*_{\text{cred}} - J^*_{\text{com}} = \begin{cases} 1/16 + 39\rho/80 > 0, & \rho \le 15/49, \\ 1/4 - \rho/8 \ge 1/8, & \rho \ge 15/49. \end{cases} \tag{8.4}$$

Proof. The first assertion is the $p < 2/3$ case of Theorem 8.1. Evaluate (8.2) at zero and (8.1) at its two states. Subtract this loss from $7/16 + 27\rho/40$ or $5/8 + \rho/16$, as appropriate in Corollary 7.2, to get (8.4). The two expressions coincide at $15/49$. Neither committed waiting rule is the discretionary minimum at $K = 0$, so the comparison genuinely uses irrevocable enforcement rather than a noncredible announcement. □

The comparison is at identical (p, ρ) and physical/legal resources. Its strict gap is asserted at $p = 1/2$, not for every probability: for example at $p = 0$ both institutions optimally choose $K = 0$.

9 The whole admitted randomized rule domain

Allow arbitrary Borel lotteries over legal authority actions, every realization satisfying (5.2) and the cash/protection constraints. Paid-in capital is public before the run choice. A date-zero lottery over it is independent of later Nature, as specified in Section 5. Hidden capital or a preliminary signal about future shocks would be a different information structure.

Theorem 9.1 (Randomization does not improve the design values)**.** *The deterministic optima in Theorems 7.1 and 8.1 are also optimal over this entire randomized rule domain. Sequentially credible authority lotteries are almost surely degenerate at their deterministic statewise minima.*

Proof. At fixed K, H, Theorem 6.1 gives a unique global minimum in each creditor history. The expected nonnegative loss gap of an optimal authority lottery must be zero. Consequently that gap is zero almost surely, which forces the unique action. This applies also to declared histories of zero equilibrium probability.

Under commitment, first fix K, H and a lottery over waiting actions that has not been revealed when the creditor decides. Replace x pointwise by $x^*(y) = [K - H + y]_+$. This is legal and weakly lowers real cost, preserving the creditor's cash $\gamma(1 - y)$. The possible y's lie in the fixed interval $[0, \min\{1, 1 + H - K\}]$. Their mean $\bar{y} = \mathbb{E}y$ lies there too, and $\bar{x} = [K - H + \bar{y}]_+$ is legal. Convexity of (6.3) gives

$$f_H(\bar{y}; K) \leq \mathbb{E} f_H(y; K).$$

In this particular formula the inequality follows directly from $\mathbb{E}y^2 \geq (\mathbb{E}y)^2$ and $\mathbb{E}[K - H + y]_+ \geq [K - H + \mathbb{E}y]_+$. The creditor's expected waiting cash remains $\gamma(1 - \mathbb{E}y)$, so its comparison to $\gamma/2$, including the declared wait-at-tie choice, is preserved. Hence the best waiting lottery cannot improve on the deterministic waiting commitment. If running is selected, its fixed resource losses cannot be undone, and an action lottery cannot improve the unique run minimum. The proof of Theorem 8.1 shows that inducing this run never beats the best feasible no-run commitment at the same K, H.

If a signal or action lottery is revealed before the creditor moves, condition on that signal. Apply the same argument to the residual unrevealed distribution; each conditional result is bounded below by the same best deterministic cost at that K, H. Taking expectations proves the result for mixtures of revealed and unrevealed randomization. All actions and losses are bounded, so these expectations exist.

Finally a public capital lottery averages losses at its realized K's and cannot be lower than their global deterministic minimum. This last expectation uses the unchanged conditional shock parameters (p, ρ): the lottery is before and independent of Nature and contains no signal about future shocks. Without that primitive the displayed averaging argument would be unjustified. □

This reduction uses the model's cash-linear creditor payoff, the convex reduced cost, public capital and specified timing. It does not assume all legal randomized policies away. Nor does it prove robustness to a different creditor equilibrium selection at indifference.

10 All-capital prices, sharp sets, and credibility

10.1 Actual repayments and the complete price functions

Take competitive risk-neutral pricing with unit discounting. Let Q_A be the price per unit face of A's designated debt, and Q_B the price of B's entire face-γ wholesale claim. Entrepreneurial surplus belongs to other owners and is not an extra cash payoff on that claim.

Proposition 10.1 (The entire discretionary price curve)**.** *For every $K, p, \rho \in [0, 1]$, selected discretionary prices are*

$$Q_A = \begin{cases} 1 - p(1 - \rho)K, & 0 \leq K < 1/2, \\ 1 - p[(1 - \rho) + \rho(K - 1/2)], & 1/2 \leq K < 1, \\ 1 - p, & K = 1, \end{cases} \tag{10.1}$$

and

$$Q_B = \begin{cases} \gamma(1-p/2), & 0 \le K < 1/2, \\ \gamma\{1-p[(1-\rho)(1-K)+\rho/2]\}, & 1/2 \le K < 1, \\ \gamma(1-p\rho/2), & K = 1. \end{cases} \tag{10.2}$$

The unconditional run probability is

$$\mathbb{P}(\text{run}) = \begin{cases} p, & 0 \le K < 1/2, \\ p\rho, & 1/2 \le K < 1, \\ 0, & K = 1. \end{cases} \tag{10.3}$$

A protected senior face $1/2$ *has price* $1/2$*, and AT1 face* $1/4$ *has price* $(1-p)/4$*.*

Proof. Below $1/2$, both states run. Their A haircuts are respectively K and zero, and each run returns half of B's face to its creditor. On $[1/2, 1)$, the isolated state waits with $(x, y) = (1, 1-K)$, while the joint state runs with $x = K - 1/2$. At $K = 1$ both states wait: A's haircut is one in each, while B's are respectively zero and one half. Weight these actual cash payments by $p(1-\rho)$ and $p\rho$ and add full normal repayment with weight $1-p$. This proves (10.1)–(10.2); it also shows why $K = 1$ must not be filled in by a continuous interpolation of the A price. The same run classification gives (10.3). Senior debt is always protected; AT1 is paid only outside stress. □

For comparison, the best conditional commitment at any fixed K has no runs. Its designated prices are

$$\begin{aligned} Q_A^{\text{com}} &= \begin{cases} 1-p[K+(1-\rho)/2], & 0 \le K \le 1/2, \\ 1-p[(1-\rho)+\rho K], & 1/2 \le K \le 1, \end{cases} \\ Q_B^{\text{com}} &= \begin{cases} \gamma(1-p/2), & 0 \le K \le 1/2, \\ \gamma\{1-p[(1-\rho)(1-K)+\rho/2]\}, & 1/2 \le K \le 1. \end{cases} \end{aligned} \tag{10.4}$$

Indeed (8.1) gives A haircuts $K + 1/2, K$ below $1/2$, and $1, K$ above it; B's are $1/2, 1/2$ below and $1-K, 1/2$ above. Taking the same state expectations proves (10.4); both pieces agree at $1/2$. Senior and AT1 prices are unchanged.

At $p = 1/2$, the discretionary values at the initially relevant capital candidates are

K	Q_A	Q_B
0	1	3/8
1/2	$(1+\rho)/2$	3/8
1	1/2	$1/2 - \rho/8$.

10.2 Identification and its exact limitations

If p is unknown, the maintained full AT1 stress loss and unit risk-neutral discounting give $p = 1 - 4Q_{\text{AT1}}$. With an unknown common state-independent discount δ, the protected senior benchmark gives $\delta = 2Q_{\text{senior}}$; divide the other prices by δ before using their formulas. State-dependent risk premia instead change physical probabilities into state-price weights, as in Theorem 4.1. They cannot be discarded when inferring physical stress frequencies or welfare.

Proposition 10.2 (Sharp structural sets from the stated experiment)**.** *Consider finitely many observed, known capital choices under common* (p, ρ) *and the declared pricing and equilibrium conventions. Each discretionary observation is generated by the displayed discretionary continuation. Each commitment observation is generated by the displayed conditionally cost-minimizing*

schedules (8.1)*; knowing only that some irrevocable commitment exists would not suffice. The sharp parameter set S is exactly the subset of $[0,1]^2$ satisfying all applicable price, run and observed state-frequency equations jointly. For that specified commitment experiment, the equations are* (10.4) *and the zero-run restriction.*

Fix the policy targets before using the observations: let $\kappa_{\mathrm{D}}(p,\rho)$ be the smallest cost-minimizing member of $\mathcal{C}(p,\rho)$ in Theorem 7.1, and let $\kappa_{\mathrm{C}}(p,\rho)$ be the least member of the complete commitment optimizer set in Theorem 8.1. For either fixed target $j \in \{\mathrm{D},\mathrm{C}\}$, its sharp identified set is $\{\kappa_j(p,\rho) : (p,\rho) \in S\}$. This targets a declared deterministic optimal capital choice, not every optimal lottery or off-path commitment rule.

Proof. Necessity follows from the actual cash and run computations. Conversely, every retained pair specifies the balance sheets and law above; the proved authority and creditor rules generate every stipulated price and behavior observation in that one model. Thus every retained parameter and its policy target are attainable. The converse uses common structural parameters across policies and claims, rather than a separate recovery model for each observation. □

Under discretion, at $K = 0$, the two debt prices, the AT1 and senior prices, and aggregate run frequency are all independent of ρ. In particular, run counts there do *not* resolve the ambiguity left by prices. At any $K \in [1/2, 1)$, known $p > 0$ and run frequency identify ρ; observed H labels among stress episodes also identify it. The displayed price slopes supply the corresponding policy-variation identification whenever their coefficient is nonzero. At $p = 0$, these realized financial data do not identify the never-realized conditional ρ, although the hypothetical all-state credibility tests remain specified.

Under discretion at $p = 1/2, K = 0$, the same whole quoted claim-price vector is therefore consistent with either optimal capital $1/2$ or 1, depending on which side of $15/49$ the unidentified ρ lies. If the declared target is worst-case incremental loss over the identified $\rho \in [0,1]$, Corollary 7.3 gives $K = 1$. This is a specified robust counterfactual, not point identification of the physical parameter.

Proposition 10.3 (Identical prices and behavior, different credibility)**.** *If the authority's operating-cost coefficient is unknown, even the complete prices, realized resolutions, run frequencies and observed stress labels just described need not identify the all-history credibility of a given announced rule.*

Proof. For this comparison only, set $K = 0$ and replace $x/4$ by cx, where $c \in \{1/4, 1/3\}$. Keep all other primitives and the common pricing kernel fixed. Both economies are legal under the same cash capacity: $b_A + b_B \le 5/4 - x/2$, so even with the advance and operating cost total disbursement is at most $7/4 + (c - 1/2)x \le 7/4 < 2$.

After a run, for either $H \ge 0$, loss has x-slope $c > 0$, so its unique optimum is zero. After waiting, first minimize x at $[y - H]_+$. The reduced cost is

$$f_{H,c}(y) = 1 + H - y + c[y - H]_+ + y^2/2.$$

Its derivative is negative through $y = H \le 1/2$, and thereafter equals $y + c - 1$. The unique feasible minimum is

$$y = 1 - c > 1/2, \qquad x = 1 - c - H.$$

For both c's and both states these actions satisfy all bounds. The creditor therefore strictly runs in every stress state in both economies. Actual resolutions have $x = 0$; all actual claim prices, run frequencies and observed state labels coincide.

Announce the same complete rule in both economies: after running choose $x = 0$; after waiting choose $y = 3/4$ and $x = 3/4 - H$. It is credible when $c = 1/4$. When $c = 1/3$ it is strictly

suboptimal at each unvisited waiting history, where the unique alternative has $y = 2/3, x = 2/3 - H$. Investors anticipate that deviation if waiting occurs; it still induces strict running, with unchanged observed repayments. Thus the same observation law is compatible with both credibility indicators. □

This comparison concerns unknown authority technology in an explicitly specified two-model class. It does not apply the numerical capital formula for $c = 1/4$ to a different c. It shows directly why expected loss allocation is not an observation of optimality at unused legal histories. Known objective primitives or informative counterfactual behavior restrictions are needed to make that inference.

11 Scope and dependencies

11.1 Scope

The exact target is (1.1) for the fixed institution, including its full legal rule set and both named institutional variants. Theorem 6.1 proves every authority comparison and the actual creditor response. Theorem 7.1 gives an exact optimum for every $(p, \rho) \in [0, 1]^2$; Theorem 8.1 solves commitment over the same entire capital interval and legal actions. Theorem 9.1 includes all the admitted lotteries. Both declared failure states and all run/wait histories are checked, including those assigned zero probability.

These results retain insured and senior claims, separate AT1 and resolution debt, equity, actual cross-bank receipts, resource-constrained recapitalization, statutory bridge operations, temporary funding and endogenous runs simultaneously. The authority's conditional objective and the ex ante welfare ledger are explicit; financial transfers and real losses are counted separately. The nonlinear production loss is derived from a stated constrained technology. Corollaries 7.2 and 8.2 exhibit an optimal instrument switch and a strict commitment cost, rather than evaluating an exogenously unique rule without an instrument.

Proposition 10.1 gives all-capital prices and actual run frequencies; Proposition 10.2 gives sharp structural sets under the stated experiment. Proposition 10.3 explicitly disproves identification of all-history credibility from the same realized prices and behavior when authority costs are unknown. It preserves the distinction between formal loss-imposition powers, market expected repayments and sequential optimality. Theorem 4.1 states the general common-kernel restrictions under which price bounds are sharp.

Theorems 2.1–3.2 supply an additional exact framework for the stated finite-history represented institutions. It treats general legal continua, actual payment feedback, correct belief consistency, pricing and budgets, and reports empty feasibility, unbounded values and nonattainment separately. The concrete institution proves attainment; the general theorem promises only the correct endpoint or feasible ϵ-optimal result.

The paper does not claim that every banking system has this particular law or cost technology, that general clearing is unique without leakage, that arbitrary infinite strategic games admit the finite encoding, or that all creditor indifference selections have the same optimal design. A different legal action or information structure must be incorporated into its own full domain. In the actual institution here none of its admitted continuous authority actions or randomized rules remains unexamined.

The prior attempts' quadratic exposure-cost and observable-risk theorems are preserved and credited, while the clearing, run and nontrivial instrument problems they expressly left unresolved are jointly solved here for the fixed institution. The original question requests explicit identification assumptions, not execution of a particular empirical dataset. No replication of

BIS data or calibration to actual resolution agencies is claimed.

11.2 External results

The substantive external mathematical ingredients are effective real quantifier elimination and algebraic sampling, as stated and used in Theorem 3.2, and the finite-game Nash theorem used in Lemma 3.1. The latter applies to a genuinely finite perturbed agent game; the remainder of the continuation and consistency argument is supplied here. The Kreps–Wilson citation credits the sequential-equilibrium definition, whose exact consistency and continuation tests were reproduced.

Use of AI tools

The author used research AI agents that the author built and configured (GPT-6 Astra max, Claude Opus 5.5 max and GPT-6 Astra Pro) during the proof work, for proof search and construction, proof checking, translation, Python computation and similar tasks. The final manuscript was written, and the proofs were checked, by the human author. All responsibility for this work rests with the author personally.

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[10] GPT 6 Astra Ultra (2026), *Credible bank resolution: Systemic states, discretionary write-downs, and risk selection*, version 2. Second attempt. Its all-state quadratic exposure-cost and interacting-risk results are credited; it expressly leaves clearing, runs and nontrivial credible instrument design unresolved.